

**TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY**



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**A Large-Scale Vietnamese News
Dataset for Text Classification:
Construction and Evaluation**

**INFORMATION TECHNOLOGY
PROJECT**

COMPUTER SCIENCE

HO CHI MINH CITY, 2026

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Instructor

Ph.D. Trần Thanh Phước

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Ho Chi Minh city, May 10, 2026

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We hereby certify that this is our own research work and is under the scientific guidance of Ph.D. Tran Thanh Phuoc. The research contents and results in this topic are honest and have not been published in any form before. The data in the tables for analysis, comments and evaluations were collected by the author himself from various sources clearly stated in the reference section.

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Ho Chi Minh City, May 10, 2026

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A Large-Scale Vietnamese News Dataset for Text Classification: Construction and Evaluation

SUMMARY

The Vietnamese language presents distinctive natural language processing (NLP) challenges, which are compounded by a critical shortage of standard benchmark datasets. To address this gap, this paper introduces BN-VN3S, a large-scale Vietnamese news dataset comprising 972,214 clean articles refined from over 1.01 million raw samples, was systematically collected from three highly reputable publishers, including VnExpress, VietNamNet, and Dân Trí. Spanning an extended publication period from 2019 to 2025 across 10 topical categories, this dataset provides unprecedented diversity. Utilizing this resource, we conduct a comprehensive empirical evaluation of ten different models across three paradigms: Traditional Machine Learning (Naïve Bayes, Logistic Regression, LinearSVC, SGDClassifier), Deep Learning (TextCNN, BiGRU, TextRCNN), and Transformers (mBERT, XLM-R, PhoBERT). Our study analyzes classification performance through four critical dimensions: model architecture, temporal data shift, source-origin bias, and training data scale. Experimental results surprisingly reveal that deep learning and traditional models surpass Transformers in overall performance; BiGRU achieved the highest macro F1-score of 92.42%, closely followed by LinearSVC at 92.33%, whereas PhoBERT reached 87.98% and mBERT lagged at 78.73%. However, under temporal distribution shifts evaluated on 2023–2025 data, Transformers particularly PhoBERT demonstrate superior robustness and maintain the most stable performance. Furthermore, we find that models are highly sensitive to source bias; BiGRU suffered a substantial performance drop of up to 9.71 F1 points during cross-source evaluation, while Naïve Bayes and mBERT were significantly more resilient. Finally, an analysis of data scale reveals a clear sample efficiency trade-off: traditional models excel in extreme low-data regimes (trained on just 1–5% of the data), whereas Transformers experience a crossover point and

dominate when trained on 10–25% or more of the data. By publicly releasing this 1-million-article dataset and its multi-model benchmark, this work provides actionable insights and practical recommendations for optimizing Vietnamese news classification systems in real-world environments.

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LIST OF ABBREVIATIONS

BERT	Bidirectional Encoder Representations from Transformers
NLP	Natural Language Processing
BiGRU	Bidirectional Gated Recurrent Unit
BoW	Bag-of-words
TF-IDF	Term Frequency - Inverse Document Frequency
TF	Term Frequency
IDF	Inverse Document Frequency
P	Probability
TextCNN	Text Convolutional Neural Network
TextRCNN	Recurrent Convolutional Neural Network for Text
SGD	Stochastic Gradient Descent
mBERT	Multilingual Bidirectional Encoder Representations from Transformers
XLM-R	Cross - lingual Language Model - RoBERTa
LinearSVC	Linear Support Vector Classifier
AUC	Area Under the Curve
LoRA	Low-Rank Adaption
PhoBERT	Pre-trained BERT for Vietnamese
XGBoost	Extreme Gradient Boosting
CNN	Convolutional Neural Network

CHƯƠNG 1. INTRODUCTION AND TOPIC OVERVIEW

1.1 Reasons for choosing the topic

With the rapid growth of online news platforms, a large amount of textual information is published every day. Automatically classifying news articles into different categories has become an important task in Natural Language Processing (NLP) to help users quickly find relevant information and improve information management systems. This project focuses on Vietnamese news classification because Vietnamese NLP resources are still relatively limited compared to other languages. By collecting news articles from popular Vietnamese online newspapers such as VnExpress, VietnamNet, and Dân Trí, this study aims to explore and compare different machine learning and deep learning approaches for classifying Vietnamese news articles, contributing to research and applications in Vietnamese text processing.

1.2 Objectives of the project

The objective of this project is to build a Vietnamese news classification system using articles collected from online newspapers such as VnExpress, VietnamNet, and Dân Trí. The project aims to preprocess and organize the collected data into different categories and apply various text classification methods to automatically classify news articles. Another goal of the project is to evaluate and compare the performance of different models, including traditional machine learning, deep learning, and transformer-based approaches, in order to identify effective techniques for Vietnamese news classification tasks.

1.3 Scope

The scope of this project focuses on building and evaluating a Vietnamese news classification system using articles collected from several major online newspapers, including VnExpress, VietnamNet, and Dân Trí. The project mainly concentrates on the processes of data collection, preprocessing, feature representation, and model training for text classification. Various approaches such as traditional machine learning, deep learning, and transformer-based models are applied to evaluate their effectiveness in classifying Vietnamese news articles into predefined categories. However, the project is limited to textual data from selected news sources and does not cover multimedia content or real-time deployment systems.

1.4 Article structure

We organize this paper into five main chapters. Chapter 1 provides an overview of the rationale, objectives, scope, and structure of the research. Chapter 2 delves into the theoretical foundations, including fundamental knowledge of natural language processing, applied machine learning, and an overview of related works that we have surveyed. Chapter 3 details the methodology we have built, including the data collection process, normalization steps, and the proposed model architecture. Chapter 4 details the experiments we have conducted, including a description of the experimental data, performance metrics, parameter configurations, results, and an in-depth discussion of those results. Finally, Chapter 5 summarizes the main results, points out the remaining limitations, and suggests future directions.

CHƯƠNG 2. THEORETICAL BACKGROUND

2.1 Text classification

Text classification is a fundamental task in Natural Language Processing (NLP) that involves automatically assigning predefined categories to textual documents. The main objective of text classification is to organize large volumes of textual data into meaningful groups based on their content. This technique has been widely applied in various domains such as spam filtering, sentiment analysis, document categorization, and news classification.

In the context of news classification, the goal is to automatically categorize news articles into specific topics such as politics, sports, business, technology, or entertainment. With the rapid growth of online news platforms and digital media, the amount of textual information available on the internet has increased significantly. Manual categorization of such large datasets is time-consuming and inefficient, making automated text classification methods essential for efficient information management.

A typical text classification system consists of several stages, including data preprocessing, text representation, and the application of classification algorithms.

2.2 Text preprocessing

Text preprocessing is an important step in natural language processing tasks. Raw text data often contains noise such as punctuation, special characters, and inconsistent formatting, which can negatively affect model performance. Therefore, preprocessing techniques are applied to clean and standardize the text before further processing.

Common preprocessing steps include converting text to lowercase, removing punctuation and special characters, eliminating stop words, and normalizing words. These steps help reduce unnecessary information and improve the quality of the textual data used for training classification models.

2.3 Text Representation

Since machine learning models cannot directly process raw textual data, text must be transformed into numerical representations. This process is known as text representation or feature extraction. Two widely used methods for representing text data are Bag-of-Words and TF-IDF.

2.3.1 Bag-of-Words

The Bag-of-Words (BoW) model is one of the simplest and most widely used approaches for representing textual data. In this method, a document is represented as a collection of words without considering grammar or word order. Each document is converted into a vector where each dimension corresponds to a unique word in the vocabulary. Here is how it work:

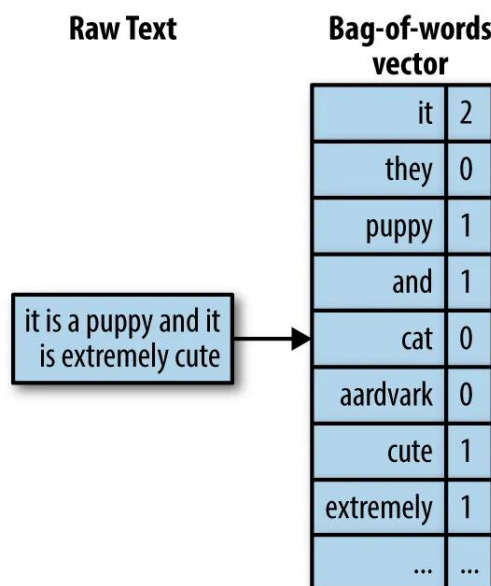


Image 1: Bag-of-words example

The value in each dimension indicates the frequency of the corresponding word in the document. Although the Bag-of-Words model is simple and computationally efficient, it ignores the semantic relationships between words and does not capture contextual information within the text.

2.3.2 Term Frequency–Inverse Document Frequency (TF-IDF)

Term Frequency–Inverse Document Frequency (TF-IDF) is an improved version of the Bag-of-Words model that assigns weights to words based on their importance within a document and across the entire dataset.

The TF component measures how frequently a word appears in a document, while the IDF component measures how unique the word is across all documents. Words that appear frequently in a specific document but rarely in other documents will receive higher TF-IDF scores.

By emphasizing informative words and reducing the influence of common words, TF-IDF provides a more meaningful representation of text data for machine learning algorithms.

TF-IDF (Term Frequency–Inverse Document Frequency) is a statistical method used in natural language processing and information retrieval to evaluate how important a word is to a document in relation to a larger collection of documents. TF-IDF combines two components:

+ Term Frequency: Measures how often a word appears in a document.

A higher frequency suggests greater importance

$$+ \text{TF}(t,d) = \frac{\text{Number of times term } t \text{ in document } d}{\text{Total number of terms in document } d}$$

- + **Inverse Document Frequency (IDF):** Reduces the weight of common words across multiple documents while increasing the weight of rare words. If a term appears in fewer documents, it is more likely to be meaningful and specific.

$$+ \text{IDF}(t,D) = \log \frac{\text{Total number of documents in corpus } D}{\text{Number of documents containing term } t}$$

2.4 Text Classification Algorithms

Various algorithms have been developed to perform text classification tasks. These algorithms can generally be categorized into three main groups: traditional machine learning methods, deep learning models, and transformer-based approaches.

2.4.1 Traditional Machine Learning Methods

Traditional machine learning algorithms have been widely used for text classification due to their simplicity and efficiency. These models typically rely on feature extraction techniques such as Bag-of-Words or TF-IDF to convert textual data into numerical vectors.

Common traditional classifiers include Naive Bayes, Logistic Regression, Linear Support Vector Classifier (LinearSVC), and Stochastic Gradient Descent (SGD) Classifier. These algorithms are capable of handling high-dimensional sparse data and are often used as baseline models in many text classification studies.

2.4.1.1 Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes' Theorem. It is widely used in text classification due to its simplicity and efficiency.

The model computes the probability of a class given an input document and assumes that features (words) are conditionally independent. Despite this strong assumption, it performs well on high-dimensional text data.

In NLP, Multinomial Naive Bayes is commonly used with Bag-of-Words or TF-IDF representations.

Advantages: fast, simple, effective with sparse data.

Limitations: independence assumption, cannot capture complex relationships.

2.4.1.2 Logistic Regression

Logistic Regression is a linear classification model that estimates the probability of a class using a logistic (sigmoid) function. It models the relationship between input features and the probability of a target class.

The probability is computed as:

$$P(C=1 | X) = \frac{1}{1 + e^{-(w^T X + b)}}$$

where w is the weight vector and b is the bias term.

For multi-class classification, the model is extended using the softmax function. Logistic Regression is particularly effective for text classification when combined with TF-IDF features.

2.4.1.3 Linear Support Vector Classifier (LinearSVC)

LinearSVC is a linear implementation of the Support Vector Machine (SVM) designed for efficient classification in high-dimensional spaces, particularly in Natural Language Processing tasks.

The model aims to find an optimal hyperplane that separates different classes by maximizing the margin between them. The decision function is defined as:

$$f(X) = w^T X + b$$

where w is the weight vector and b is the bias term.

Unlike standard SVM implementations, LinearSVC is optimized for large-scale linear classification and is more suitable for sparse datasets such as text represented by TF-IDF features. It uses a linear kernel and is implemented efficiently using libraries such as LIBLINEAR.

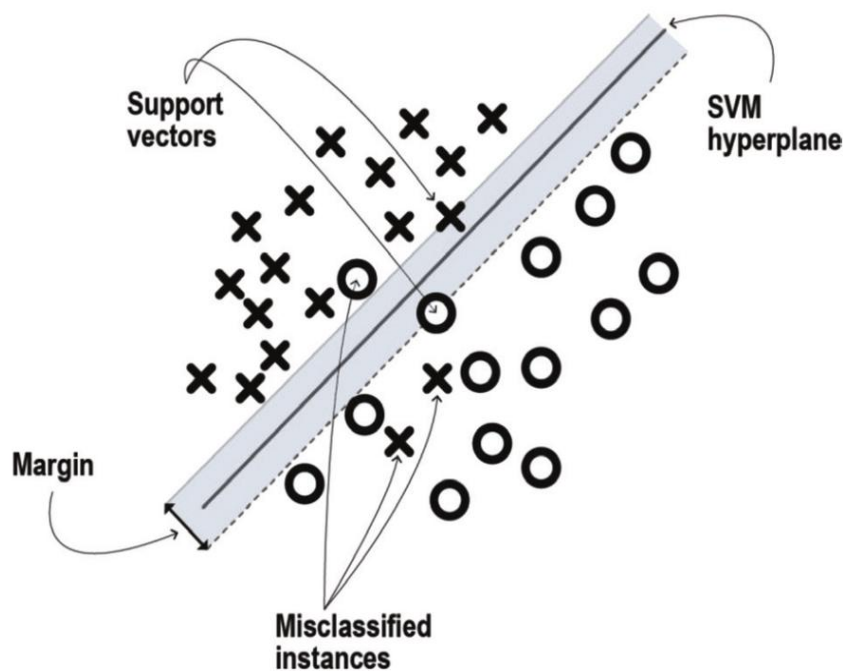


Image 2: LinearSVC

2.4.1.4 Stochastic Gradient Descent (SGD) Classifier

The SGD Classifier is a linear classification algorithm that aims to learn an optimal decision boundary (hyperplane) to separate data points from different classes. It iteratively updates model parameters to minimize a loss function, typically cross-entropy or hinge loss, using Stochastic Gradient Descent.

Unlike traditional optimization methods, SGD updates the model parameters incrementally using individual samples or small mini-batches. This makes it highly efficient and scalable for large datasets.

How it Differs from Other Classifiers

Stochastic Optimization:

Unlike batch gradient descent, which processes the entire dataset at once, SGD updates parameters using one sample at a time. This significantly reduces computational cost and enables faster convergence on large-scale data.

Linearity:

The SGD Classifier is a linear model that constructs a linear decision boundary. While this limits its ability to model complex relationships, it performs well in high-dimensional spaces such as text data in Natural Language Processing.

Regularization:

SGD supports L1 and L2 regularization, which helps prevent overfitting by penalizing large parameter values. This is especially important when working with sparse and high-dimensional feature representations like TF-IDF.

2.4.2 Deep Learning Models

Deep learning approaches have gained significant attention in recent years for natural language processing tasks. Unlike traditional methods that rely on manually engineered features, deep learning models can automatically learn complex representations from text data.

Models such as TextCNN, BiGRU, and TextRCNN have been widely applied for document classification. These models use neural network architectures to capture contextual information and semantic relationships within textual data, leading to improved classification performance.

2.4.2.1 TextCNN

TextCNN is a convolutional neural network architecture adapted for text classification tasks. It applies one-dimensional convolutional filters over word embeddings to capture local features such as n-grams within a sentence.

The model typically consists of an embedding layer followed by multiple convolutional layers with different filter sizes, allowing it to extract features at various granularities. A max-pooling layer is then applied to select the most important features, followed by a fully connected layer for classification.

TextCNN is particularly effective in capturing local semantic patterns and is computationally efficient compared to recurrent models. Here are the overall architecture of TextCNN

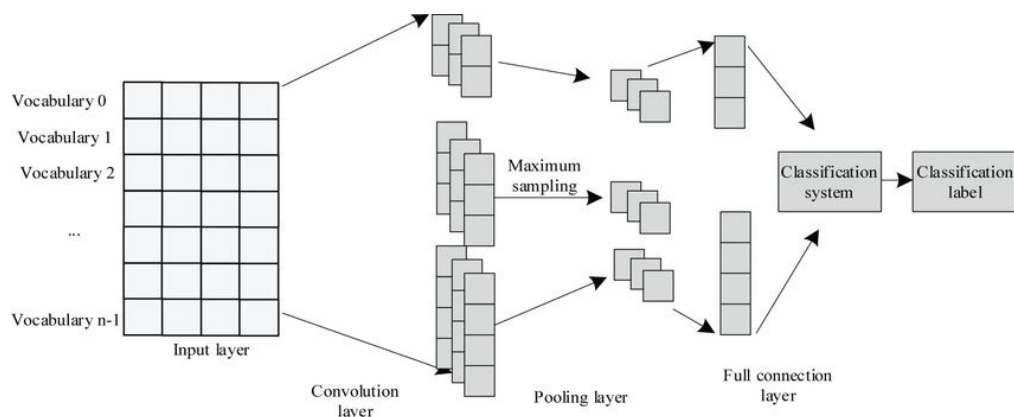


Image 3: TextCNN architecture

Advantages:

- Captures local contextual features (n-grams) effectively.
- Fast training and inference.
- Simple architecture with strong performance.

Limitations:

- Limited ability to capture long-range dependencies.
- Fixed-size context window.

2.4.2.2 Bidirectional Gated Recurrent Unit (BiGRU)

BiGRU is an extension of the Gated Recurrent Unit (GRU) that processes input sequences in both forward and backward directions. This bidirectional structure enables the model to capture contextual information from both past and future tokens in a sequence.

GRU units use gating mechanisms (reset gate and update gate) to control the flow of information and address the vanishing gradient problem found in traditional recurrent neural networks.

By combining forward and backward representations, BiGRU produces richer contextual embeddings, making it suitable for tasks where word order and context are crucial.

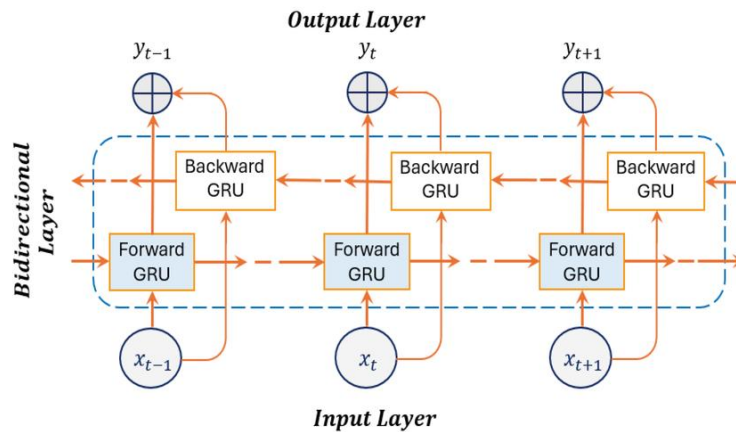


Image 4: BiGRU architecture

Advantages:

- Captures sequential dependencies effectively.
- Considers both past and future context.
- More efficient than LSTM.

Limitations:

- Slower training compared to CNN.
- Difficult to parallelize.

2.4.2.3 TextRCNN (Recurrent Convolutional Neural Network)

TextRCNN is a hybrid architecture that combines recurrent neural networks and convolutional neural networks to leverage the strengths of both approaches.

The model first uses a recurrent layer (e.g., RNN, LSTM, or GRU) to capture contextual information across the entire sequence. It then applies a convolutional or pooling operation to extract the most informative features for classification.

This combination allows TextRCNN to capture both long-range dependencies and local semantic features, making it more expressive than using CNN or RNN alone.

Advantages:

- Captures both global context and local features.
- Better representation of text semantics.
- Improved performance on complex classification tasks.

Limitations:

- Higher computational cost, more complex architecture.
- Longer training time.

2.4.3 Transformer-Based Models

Transformer-based architectures represent the state-of-the-art in many tasks within Natural Language Processing. These models utilize self-attention mechanisms to capture long-range dependencies and contextual relationships between words in a sequence. Pretrained Transformer models are typically trained on large-scale corpora and then fine-tuned for specific downstream tasks such as text classification.

2.4.3.1 Multilingual BERT (mBERT)

mBERT is a multilingual version of BERT trained on text from multiple languages using a shared vocabulary. It enables cross-lingual understanding and can be applied to tasks involving multiple languages without requiring language-specific models.

The model uses bidirectional encoding, allowing it to capture context from both left and right directions simultaneously. The image below will show the overall architecture of this model.

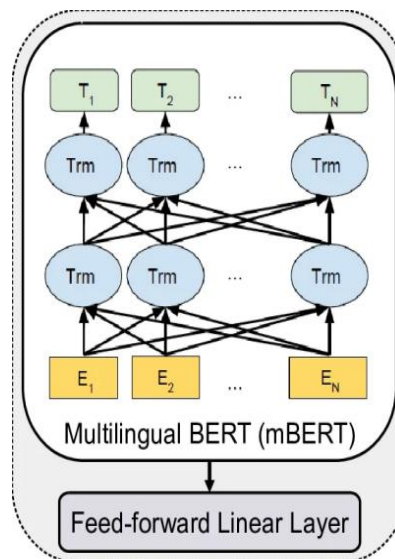


Image 5: Multilingual BERT architecture

Advantages:

- Supports multiple languages
- Effective for cross-lingual tasks
- No need for separate models per language

Limitations:

- Performance may be lower than monolingual models
- Limited optimization for specific languages

2.4.3.2 XLM-R (Cross-lingual RoBERTa)

XLM-R is a robust multilingual Transformer model trained on a large-scale multilingual dataset. It improves upon mBERT by using better training strategies and larger training data.

XLM-R is designed to handle cross-lingual tasks more effectively and provides strong performance across a wide range of languages.

Advantages:

- Strong cross-lingual performance
- Trained on large-scale multilingual data
- Better generalization than mBERT

Limitations:

- Requires significant computational resources
- Larger model size

2.4.3.3 PhoBERT

PhoBERT is a Transformer-based model specifically designed for Vietnamese language processing. It is pretrained on large-scale Vietnamese corpora and optimized for Vietnamese linguistic characteristics.

PhoBERT has demonstrated state-of-the-art performance on various Vietnamese NLP tasks, including text classification, named entity recognition, and part-of-speech tagging.

Advantages:

- Optimized for Vietnamese language
- High performance on Vietnamese NLP tasks
- Captures linguistic characteristics effectively

Limitations:

- Limited to Vietnamese language

2.5 Applications of these models

Text classification has many practical applications in real-world systems. One common application is news categorization, where news articles are automatically classified into predefined categories such as politics, sports, technology, or entertainment. This helps users quickly find relevant information and improves the organization of large news datasets.

In this project, the classification models are applied to automatically categorize news articles based on their textual content. By analyzing the words and patterns within the documents, the system can predict the appropriate category for each article. This approach can significantly reduce the manual effort required to organize large collections of news data.

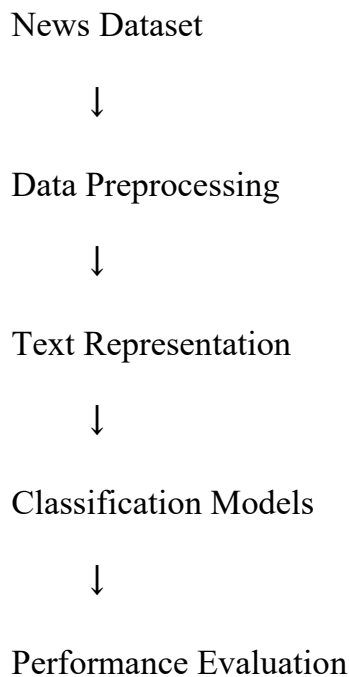
CHƯƠNG 3. PROPOSED MODEL

3.1 System Overview

This chapter presents the proposed methodology for analyzing factors that influence the performance of Vietnamese news classification systems. The proposed framework aims to evaluate how different experimental conditions affect the classification accuracy of various machine learning and deep learning models.

The overall system consists of several stages, including data collection, preprocessing, text representation, classification modeling, and evaluation. Each stage plays an important role in transforming raw news articles into structured representations that can be processed by classification algorithms.

The general workflow of the proposed system is illustrated as follows:



First, news articles are collected from multiple sources to construct the dataset used in this study. The collected text data is then cleaned and preprocessed to remove unnecessary information and standardize the format of the documents.

After preprocessing, the textual data is transformed into numerical feature vectors using text representation techniques. These features are then used as inputs to classification models that categorize the news articles into predefined topics. Finally, the performance of the models is evaluated using standard evaluation metrics.

3.2 Data Collection

The dataset used in this study consists of Vietnamese news articles collected from multiple online news sources. These articles cover a variety of topics such as politics, sports, technology, entertainment, and business.

Collecting news articles from different publishers allows the study to analyze the impact of news sources on classification performance. Different news websites may use different writing styles, vocabularies, and article structures, which may affect how machine learning models learn patterns from the data.

In addition to the source of the news articles, the dataset also includes articles from different time periods. This allows the study to investigate whether temporal variations in news content influence classification accuracy.

The collected dataset is organized into predefined categories representing different news topics. These labeled articles serve as the input data for training and evaluating the classification models.

3.3 Data Preprocessing

Raw textual data collected from online news sources often contains noise such as punctuation, special characters, HTML tags, and inconsistent formatting. Therefore, preprocessing is an important step to ensure that the data is clean and suitable for machine learning models.

The preprocessing stage includes several operations to standardize and clean the text data. First, all text is converted into lowercase to ensure consistency in word representation. Next, punctuation marks and special characters are removed because they do not contribute meaningful information for text classification.

Stopwords, which are commonly occurring words with little semantic meaning, may also be removed to reduce noise in the dataset. Additionally, tokenization is applied to divide the text into individual words or tokens that can be processed by machine learning algorithms.

These preprocessing steps help improve the quality of the dataset and enhance the effectiveness of the subsequent feature extraction process.

3.4 Text Representation

Since machine learning algorithms cannot directly process raw text, the textual data must be converted into numerical feature vectors. This process is known as text representation or feature extraction.

In this study, text representation techniques are used to transform news articles into structured numerical vectors that capture the importance of words within each document. These vectors serve as input features for classification models.

One commonly used method for text representation is the Term Frequency–Inverse Document Frequency (TF-IDF) technique. TF-IDF assigns weights to words based on how frequently they appear in a document and how unique they are across the entire dataset. Words that appear frequently in a specific document but rarely in other documents receive higher weights.

By converting text into TF-IDF vectors, the proposed system can effectively represent textual information in a numerical form that enables machine learning algorithms to identify patterns and relationships between words and document categories.

3.5 Classification Models

To evaluate the performance of different approaches for Vietnamese news classification, several classification models are applied in this study. These models belong to different categories of text classification techniques, including traditional machine learning methods, deep learning models, and transformer-based architectures.

Traditional machine learning models such as Naive Bayes, Logistic Regression, Support Vector Machine, and Stochastic Gradient Descent classifiers are commonly used in text classification tasks due to their simplicity and efficiency in handling high-dimensional textual data.

Deep learning models such as TextCNN, BiGRU, and TextRCNN are also included in the experiments. These models can automatically learn complex semantic patterns from textual data and capture contextual information within sentences.

In addition, transformer-based models including mBERT, XLM-R, and PhoBERT are utilized to leverage pretrained language representations. These models have demonstrated strong performance in various natural language processing tasks, particularly in multilingual and Vietnamese language processing.

By applying different classification models, the study aims to compare their performance and analyze how model architecture influences classification accuracy.

3.6 Experimental Design

The experimental design of this study focuses on analyzing four key factors that may influence the performance of Vietnamese news classification systems.

The first factor is the news source. News articles collected from different publishers may vary in writing style, vocabulary usage, and content structure. Therefore, the experiments analyze how classification models perform when trained on datasets from different news sources.

The second factor is the time period of the news articles. News content evolves over time as new events, topics, and terminology emerge. The experiments investigate whether models trained on data from different time periods exhibit variations in classification performance.

The third factor is the training dataset size. Machine learning models often require sufficient training data to learn meaningful patterns. To analyze the impact of training data size, the experiments are conducted using different proportions of the dataset. This allows the study to observe how classification accuracy changes as the amount of training data increases.

The final factor is the model architecture. Different types of classification models may have different capabilities in capturing textual patterns and contextual relationships. By comparing the performance of traditional machine learning models, deep learning models, and transformer-based models, the study aims to determine which approach is most suitable for Vietnamese news classification.

3.7 Evaluation Metrics

To measure the performance of the classification models, several evaluation metrics are used in this study. These metrics provide quantitative measures of how well the models classify news articles into their correct categories.

The most commonly used metric is accuracy, which measures the proportion of correctly classified instances among all test samples.

In addition to accuracy, other metrics such as precision, recall, and F1-score are also used to provide a more comprehensive evaluation of model performance. Precision measures the proportion of correctly predicted positive instances among all predicted positive instances. Recall measures the proportion of correctly predicted positive instances among all actual positive instances. The F1-score is the harmonic mean of precision and recall and provides a balanced evaluation of classification performance.

These evaluation metrics allow the study to systematically compare different models and analyze how various factors influence classification accuracy.

CHƯƠNG 4. Methodology

4.1 Problem Statement and Motivation

News classification is an important task in natural language processing that aims to automatically categorize news articles into predefined topics such as politics, sports, technology, business, and entertainment. With the rapid growth of digital media and online news platforms, a large number of news articles are published every day. Manually organizing and categorizing this massive amount of information is inefficient and time-consuming. Therefore, automated news classification systems have become essential for improving information management and retrieval.

Machine learning and deep learning techniques have been widely applied to news classification tasks due to their ability to automatically learn patterns from textual data. By analyzing the content of news articles, these models can assign appropriate categories to documents with a certain level of accuracy. However, the performance of news classification models can be influenced by several factors related to both the dataset and the modeling approach.

In this study, we aim to design a news classification model that can effectively analyze Vietnamese news articles and learn meaningful representations from textual data collected from multiple news sources. We adopt a machine learning-based approach due to its ability to handle high-dimensional text features and its proven effectiveness in document classification tasks. Our objective is to develop a multi-class classification model that maps news article content to corresponding topic categories such as politics, sports, technology, and entertainment, thereby supporting efficient organization and retrieval of news information. This chapter presents the overall modeling framework, including data collection, preprocessing, text representation, classification models, and evaluation

strategies, with an emphasis on methodological consistency and experimental analysis.

4.2 Data Loading and Structural Normalization

The dataset used in this study was collected from multiple Vietnamese online news platforms, including Dan Tri, VnExpress, and VietnamNet. Since there is no publicly available dataset that sufficiently covers Vietnamese news classification tasks, data collection was performed directly from these sources.

Due to differences in website structures, the raw data contains inconsistencies such as HTML tags, advertisements, navigation elements, and other irrelevant content. Therefore, preprocessing steps were applied to extract the main textual content and remove non-informative elements.

Each article was transformed into a standardized format, including:

- Title
- Content
- Category label
- Publication time

This structural normalization ensures consistency across all data samples, making the dataset more suitable for subsequent processing steps such as feature extraction and model training.

4.3 Data Cleaning and Quality Assurance

Based on the results obtained from the exploratory data analysis (EDA), several data cleaning procedures were performed to improve the quality, consistency, and reliability of the dataset. These steps are essential to reduce noise, eliminate redundancy, and ensure that the data is suitable for subsequent feature extraction and model training processes.

4.3.1 Handling Missing Values

The EDA results indicate that the dataset contains a small number of missing values in the *title* field (1,672 records), while no missing values were observed in the *text* field.

Since the *title* provides important contextual information for text classification, records with missing titles were considered incomplete. Therefore, these records were removed from the dataset to maintain data integrity and avoid introducing noise into the training process.

```
--- Missing values ---  
Title NaN: 1672  
Text NaN : 0
```

Image 6: Show missing value

4.3.2 Handling Empty Strings

The analysis confirmed that there are no empty strings in either the *title* or *text* fields. This indicates that all records contain valid textual content.

As a result, no additional preprocessing steps were required for handling empty values.

```
--- Empty strings ---  
Title rỗng: 0  
Text rỗng : 0
```

Image 7: Show empty string

4.3.3 Removing Duplicate Data

A significant number of duplicate entries were identified during the EDA phase, including duplicate titles, duplicate texts, and duplicate title-text pairs. In particular, 40,989 records were found to have identical combinations of title and text.

Duplicate data can lead to biased model learning, as repeated samples may disproportionately influence the training process. To address this issue, duplicate records based on the combination of *title* and *text* were removed, ensuring that each data instance in the dataset is unique.

```
--- Duplicate ---
Title trùng: 49843
Text trùng : 42942
Title + Text trùng: 40989
```

Image 8: Show duplicate

4.3.4 Final Dataset

After applying the cleaning steps, the dataset becomes more consistent and free of redundancy. This ensures better data quality for subsequent processing and model training.

Category	#Articles	Avg.words	Min words	Max words	Unique words	Train	Dev	Test
Politics	138,969	223.29	11	16,423	720,308	111,175	13,896	13,898
Sports	127,430	196.58	10	2,232	860,568	101,944	12,743	12,743
Lifestyle	56,519	254.13	8	2,245	807,427	45,215	5,651	5,653
Business	126,567	273.14	9	4,335	981,760	101,253	12,656	12,658
Education	63,200	242.90	2	4,197	564,807	50,560	6,320	6,320
Health	100,021	243.12	11	6,634	740,728	80,016	10,002	10,003
Entertainment	99,790	224.25	5	2,243	1,230,085	79,832	9,979	9,979
Law	69,388	181.05	18	1,407	489,642	55,510	6,938	6,940
Automotive	49,540	222.57	5	2,142	407,867	39,632	4,954	4,954
World	140,790	198.06	9	2,778	990,187	112,632	14,079	14,079
Total	972,214	225.91	2	16,423	7,793,379	777,769	97,218	97,227

Note: #Articles: number of articles by category, Avg.words / Min / Max: article length statistics, Total: total of all categories.

Image 9: Final Dataset

4.4 Text Preprocessing

4.4.1 Text Cleaning

After obtaining the final dataset, the textual data still contains various types of noise such as punctuation, numerical values, inconsistent capitalization, and redundant whitespace, which may negatively affect the performance of machine learning models. To address this issue, a text cleaning process is applied to normalize and refine the data. Specifically, all text is first converted to lowercase to ensure consistency across the dataset. Then, punctuation and numerical characters are removed using regular expressions, followed by the elimination of extra whitespace to standardize the text format. In addition, Vietnamese stopwords are removed based on a predefined list loaded from an external file. These stopwords are high-frequency words that contribute little to the semantic meaning of the text, and their removal helps reduce noise, decrease the vocabulary size, and improve the quality of input features. The implementation of this process is illustrated through a function that performs these transformations sequentially. For example, the sentence “Tôi đang học về trí tuệ nhân tạo!!!” is transformed into “học trí tuệ nhân tạo” after cleaning. Overall, this step ensures that the textual data is consistent, noise-free, and suitable for subsequent processing tasks.

4.4.2 Vietnamese Word Tokenization

After the cleaning step, the text is normalized but still not segmented into meaningful linguistic units. In Vietnamese, a word may consist of multiple syllables separated by spaces, making simple tokenization methods such as whitespace splitting ineffective. To accurately segment the text, the Underthesea library is employed for Vietnamese word tokenization. This approach leverages linguistic rules and contextual information to identify correct word boundaries, allowing compound words to be preserved and represented using underscores (e.g., “trí_tuệ”, “nhân_tạo”). To improve processing efficiency, especially for large-scale datasets, multiprocessing is applied to perform tokenization in parallel across multiple CPU

cores, significantly reducing execution time compared to sequential processing. The implementation involves defining a tokenization function and applying it to the dataset in parallel. For instance, the cleaned text “học trí tuệ nhân tạo” is transformed into “học trí_tuệ nhân_tạo” after tokenization. This step plays a crucial role in preserving semantic meaning and improving the quality of text representation, thereby enhancing the performance of downstream tasks such as feature extraction and model training.

This is the dataframe after processing

	content	label	source
0	trung_quốc kêu_gọi trẻ hiến tinh_trùng zhong m...	10	VNExpress
1	hoa_hồng vợ_chồng việt_đức vợ_chồng phươg_quỳ...	10	VNExpress
2	tình chân nhảy_ukraine nga công_chiếm thành_ph...	6	VNExpress
3	đặng_ngân_đoạt hậu hoa_hậu_siêu quốc_gia thí_s...	2	VNExpress
4	vợ_chồng quan_hệ bao_nhiều tuần_hải đời_sống t...	10	VNExpress
...	...	...	...
972209	tình_yêu chàng trai vợ almeda hardwich maryvil...	10	VNExpress
972210	tp hcm chuẩn giường_trị covid ổ dịch phức_tạp ...	5	VNExpress
972211	mobifone tổ_chức giải đấu esports sinh_viên mo...	1	VNExpress
972212	đại_học ngân_hàng tp hcm tân_hiệu_trưởng ngoái...	1	VNExpress
972213	áp_lực thực_hiện trường_học hạnh_phúc tọa_đàm ...	1	VNExpress

972214 rows × 3 columns

Image 10: Text tokenize

4.5 Data Splitting and Preparation

The dataset is divided into three subsets: training, development (validation), and testing sets using a stratified splitting strategy to preserve the distribution of categories across all subsets.

As shown in image below, the dataset contains a total of 946,696 articles spanning 10 categories. The splitting ratio is approximately 80% for training, 10%

for validation, and 10% for testing. This ratio is applied consistently across all categories to ensure that each subset reflects the overall data distribution.

A closer examination of the data reveals that the distribution across categories is not uniform. For instance, the *World* and *Politics* categories contain the largest number of samples, with 137,909 and 137,475 articles, respectively. In contrast, categories such as *Automotive* (47,310 articles) and *Lifestyle* (54,794 articles) have significantly fewer samples. This imbalance may influence model performance, as categories with more data tend to be better learned by the model.

The stratified splitting ensures that this imbalance is preserved proportionally in each subset. For example, in the *Politics* category, 109,980 samples are allocated to the training set, while 13,747 and 13,748 samples are used for validation and testing, respectively. Similar proportional splits are applied to all other categories, preventing distribution shift between training and evaluation phases.

In terms of text characteristics, the dataset shows notable variation in document length. The average number of words per article ranges from approximately 182 words (*Law*) to 274 words (*Business*). The maximum length reaches up to 16,423 words, indicating the presence of extremely long documents. To ensure data consistency and reduce noise, a minimum threshold of 50 words is applied during preprocessing.

Additionally, the dataset contains a large and diverse vocabulary, with a total of 7,775,020 unique words. This richness in vocabulary increases the complexity of the classification task, as the model must learn to generalize across a wide range of linguistic patterns.

Overall, the chosen splitting strategy ensures that the training, validation, and testing sets are both representative and balanced. This setup provides a reliable

foundation for model training and evaluation while minimizing potential bias caused by uneven data distribution.

Category	#Articles	Avg.words	Min words	Max words	Unique words	Train	Dev	Test
Politics	137,475	225.36	50	16,423	719,229	109,980	13,747	13,748
Sports	124,393	200.57	50	2,232	859,468	99,514	12,439	12,440
Lifestyle	54,794	261.16	50	2,245	805,730	43,835	5,479	5,480
Business	126,006	274.22	50	4,335	981,351	100,805	12,600	12,601
Education	59,796	255.45	50	4,197	563,031	47,837	5,979	5,980
Health	99,430	244.37	50	6,634	740,240	79,544	9,943	9,943
Entertainment	90,791	244.03	50	2,243	1,223,072	72,633	9,079	9,079
Law	68,792	182.23	50	1,407	488,742	55,034	6,879	6,879
Automotive	47,310	231.71	50	2,142	406,442	37,848	4,731	4,731
World	137,909	201.52	50	2,778	987,715	110,327	13,791	13,791
Total	946,696	232.10	50	16,423	7,775,020	757,357	94,667	94,672

Note: #Articles: number of articles by category, Avg.words / Min / Max: article length statistics, Total: total of all categories.

Image 11: Data splitting

4.6 Text Representation

To enable machine learning and deep learning models to process textual data, all input text must be converted into numerical representations. In this study, three different text representation approaches are employed, including transformer-based representation, TF-IDF representation, and sequence-based representation for deep learning models.

For transformer-based models, pretrained tokenizers associated with architectures such as XLM-RoBERTa and Multilingual BERT are utilized. These tokenizers segment text into subword units and convert them into token IDs based on a predefined vocabulary. This approach allows the model to effectively handle rare and unseen words while preserving contextual meaning. All sequences are truncated to a maximum length of 128 tokens, and dynamic padding is applied during training to ensure consistent input dimensions.

In addition, TF-IDF is used as a traditional feature extraction method to establish baseline models. The vectorizer is configured with a maximum vocabulary size of 100,000 features and incorporates both unigrams and bigrams to capture contextual patterns. To reduce noise, terms that appear too rarely or too frequently

are filtered using minimum and maximum document frequency thresholds. Sublinear term frequency scaling is also applied to normalize the influence of high-frequency terms.

For deep learning models such as CNN and LSTM, a sequence-based representation is adopted. A tokenizer is used to build a vocabulary from the training data, and each document is converted into a sequence of integer indices. These sequences are then padded or truncated to a fixed maximum length based on the distribution of text lengths. This representation enables neural networks to learn semantic patterns directly from word embeddings.

4.7 Model Training

4.7.1 Traditional Machine Learning Models

Traditional machine learning models are employed as baseline approaches using TF-IDF features. These models include Multinomial Naive Bayes, Logistic Regression, Linear Support Vector Machine (LinearSVC), Random Forest, and XGBoost.

Multinomial Naive Bayes is a probabilistic model that assumes feature independence and is particularly effective for text classification tasks with word frequency features. Logistic Regression is a linear model that estimates class probabilities and performs well in high-dimensional spaces such as TF-IDF representations. LinearSVC is a support vector machine variant optimized for linear decision boundaries, which is highly efficient for sparse text data.

In addition, tree-based models such as Random Forest and XGBoost are also evaluated. These models are capable of capturing non-linear relationships in the data, although they may not perform as efficiently with sparse high-dimensional features compared to linear models. Class weighting is applied where applicable to mitigate class imbalance.

4.7.2 Deep Learning Models

Deep learning models are utilized to capture semantic and contextual patterns beyond traditional feature-based approaches. Three architectures are implemented in this study: TextCNN, Bidirectional LSTM (BiLSTM), and a hybrid TextCNN + BiLSTM model.

The TextCNN model applies multiple convolutional filters with varying kernel sizes to extract local n-gram features from text. These features are then aggregated באמצעות max-pooling and passed through a fully connected layer for classification.

The BiLSTM model processes input sequences in both forward and backward directions, allowing it to capture long-range dependencies and contextual relationships within the text. The hidden states from both directions are concatenated to form a comprehensive representation of the input.

To combine the strengths of both architectures, a hybrid TextCNN + BiLSTM model is introduced. This model leverages convolutional layers for local feature extraction and recurrent layers for sequential modeling, enabling it to capture both short-term and long-term dependencies.

All models utilize an embedding layer to transform token indices into dense vector representations. Dropout is applied to reduce overfitting, and the models are trained using the Adam optimizer with cross-entropy loss. A learning rate scheduler and early stopping mechanism are also employed to improve training stability and efficiency.

4.7.3 Transformer-based Models

Transformer-based models are adopted as the most advanced approach for text classification in this study. Pretrained multilingual models such as XLM-RoBERTa and Multilingual BERT are fine-tuned on the Vietnamese news classification task.

These models are based on the transformer architecture, which utilizes self-attention mechanisms to capture contextual relationships between words in a sequence. Unlike traditional and deep learning models, transformer-based models can effectively model long-range dependencies and complex semantic structures.

To improve training efficiency and reduce computational cost, Low-Rank Adaptation (LoRA) is applied. This technique allows only a small number of additional parameters to be trained while keeping the majority of the pretrained model fixed. As a result, the model can adapt to the target task with reduced memory usage and faster training.

Overall, transformer-based models provide superior performance due to their ability to leverage large-scale pretrained knowledge and deep contextual understanding.

To improve training efficiency and reduce computational cost, Low-Rank Adaptation (LoRA) is applied during the fine-tuning process. Instead of updating all parameters of the pretrained model, LoRA introduces a small number of trainable low-rank matrices while keeping the original model weights frozen.

This approach significantly reduces the number of trainable parameters, leading to faster training and lower memory consumption. At the same time, it preserves the knowledge learned from large-scale pretraining, allowing the model to achieve strong performance with limited computational resources.

4.8 Evaluation Metrics

To comprehensively evaluate model performance, multiple evaluation metrics were employed:

- **Accuracy:** Measures the proportion of correctly predicted samples over the total number of samples.

- Precision: Evaluates the proportion of correctly predicted positive instances among all predicted positives.
- Recall: Measures the ability of the model to correctly identify all relevant instances.
- Macro F1-score: The harmonic mean of precision and recall, averaged equally across all classes. This metric is particularly useful for imbalanced datasets.
- AUC (Area Under the ROC Curve): Measures the model's ability to distinguish between classes across different decision thresholds.

In addition to overall metrics, a detailed per-class classification report was generated, providing precision, recall, F1-score, and support for each class. This allows for deeper analysis of model performance across different categories.

4.9 Summary

This chapter presented the overall methodology for Vietnamese news classification, including data collection, preprocessing, feature extraction, model training, and evaluation.

Text data was processed and transformed using TF-IDF for traditional models, while Transformer-based models such as XLM-RoBERTa and mBERT were fine-tuned using LoRA.

Model performance was evaluated using multiple metrics to ensure a comprehensive assessment.

Overall, the proposed framework provides an effective pipeline for large-scale Vietnamese news classification and serves as the basis for the experimental results in the next chapter.

CHƯƠNG 5. EXPERIMENT

5.1 Experimental Setup

All experiments were conducted on Google Colaboratory using a single NVIDIA Tesla T4 GPU (12GB VRAM) with 51GB system RAM.

For traditional machine learning models, including Naïve Bayes, Logistic Regression, LinearSVC, and SGDClassifier, text data was represented using TF-IDF with a maximum vocabulary size of 100,000 tokens. Both unigram and bigram features were incorporated. These models were implemented using scikit-learn.

For deep learning models, including TextCNN, BiGRU, and TextRCNN, 300-dimensional pre-trained Vietnamese word embeddings were used. The models were trained with a batch size of 64 for up to 10 epochs, with early stopping (patience = 3) based on the development set F1-score. These models were implemented using PyTorch.

For Transformer-based models, including PhoBERT, mBERT, and XLM-R, full fine-tuning was performed for 3 epochs using a learning rate of 2×10^{-5} , a batch size of 32, and a maximum sequence length of 256 tokens. These models were implemented using the Hugging Face Transformers framework.

To improve training efficiency, the LoRA (Low-Rank Adaptation) technique was applied to the Transformer models. Instead of updating all model parameters during fine-tuning, LoRA introduces additional low-rank matrices into selected layers of the model, significantly reducing the number of trainable parameters while preserving model performance.

In this study, LoRA was configured with a rank $r=8$, scaling factor $\alpha=16$, and dropout rate of 0.1. Only the LoRA parameters were updated during training, while the original pretrained weights remained frozen. This approach reduces GPU

memory usage and accelerates training, making it suitable for large-scale models in limited-resource environments.

```
lora_config = LoraConfig(  
    task_type=TaskType.SEQ_CLS,  
    r=8,  
    lora_alpha=16,  
    lora_dropout=0.1,  
    bias="none"  
)
```

Image 12: LoRA configuration

All models were evaluated using macro-averaged F1-score, precision, recall, and accuracy to ensure a fair comparison across the imbalanced 10-category label distribution. Hyperparameter configurations are summarized in Fig. 4.

5.2 Evaluation Metrics

The classification performance was evaluated using four metrics: accuracy, precision, recall, and macro F1-score.

Accuracy measures the proportion of correctly predicted samples over the total number of samples. However, it may not be reliable in imbalanced datasets, as a model predicting only the majority class can still achieve high accuracy.

Precision represents the proportion of correctly predicted positive samples among all predicted positives, reflecting how accurately the model identifies positive instances.

Recall measures the proportion of correctly predicted positive samples among all actual positive samples, indicating the model's ability to capture relevant instances.

The macro F1-score is calculated as the unweighted average of F1-scores across all classes, where each F1-score is the harmonic mean of precision and recall. This metric treats all classes equally regardless of their size, making it particularly suitable for multi-class classification with imbalanced data.

5.3 Results and Comparison

Model	AUC	Accuracy	Macro-F1	Recall	Precision
Naive Bayes	98.83	87.06	86.36	87.09	86.03
Logistic Regression	99.55	91.92	91.36	92.04	90.83
LinearSVC	99.55	92.92	92.33	92.31	92.38
SGDClassifier	98.95	87.73	87.03	87.68	86.55
TextCNN	99.41	91.91	91.27	91.23	91.32
<u>BiGRU</u>	<u>99.61</u>	<u>92.43</u>	<u>92.42</u>	<u>92.47</u>	<u>92.43</u>
TextRCNN	99.59	92.12	92.16	92.20	92.12
mBERT	97.04	78.10	78.73	80.30	78.10
XLM-R	97.94	80.80	81.60	82.78	80.80
PhoBERT	99.01	86.10	87.98	87.98	86.10

Table 1: Results

Table presents the overall classification performance of all evaluated models.

Among all models, BiGRU achieved the best performance with a macro F1-score of 92.42%, making it the most effective model in this study. It also maintained a strong balance between precision (92.43%) and recall (92.47%), indicating stable and consistent classification across all classes.

Closely following BiGRU, LinearSVC obtained a macro F1-score of 92.33%, showing nearly equivalent performance. This result highlights that a traditional machine learning model combined with TF-IDF features can achieve performance comparable to advanced deep learning approaches.

TextRCNN also demonstrated competitive results with a macro F1-score of 92.16%, confirming that hybrid architectures can effectively capture both local and contextual information in text data.

On the other hand, Transformer-based models such as mBERT and XLM-RoBERTa showed noticeably lower performance, with macro F1-scores of 78.73% and 81.60%, respectively. Even PhoBERT, which is specifically pretrained for Vietnamese, achieved only 87.98%, still significantly lower than the top-performing models.

These results indicate that while Transformer models are powerful, they did not outperform simpler or more specialized models under the current experimental setup.

5.4 Discussion

RQ1: Impact of Model Architecture on Classification Performance

The experimental results show that model architecture has a significant impact on classification performance. Among all models, BiGRU achieved the best performance with the highest macro F1-score, indicating the effectiveness of recurrent architectures in capturing contextual dependencies in text.

Interestingly, LinearSVC also achieved competitive performance, demonstrating that traditional machine learning models combined with TF-IDF can still be highly effective for text classification. This suggests that feature representation plays a crucial role alongside model architecture.

In contrast, Transformer-based models such as mBERT and XLM-RoBERTa did not perform as well as expected. This indicates that advanced architectures do not always guarantee superior performance, especially under limited training configurations.

RQ2: Impact of Publication Time on Classification Performance

To investigate temporal distribution shift, we trained all models on articles published between 2019 and 2022 and evaluated them on articles from 2023, 2024, and 2025 respectively. Image **XX** shows the macro F1 performance of each model across evaluation years.

The results indicate a clear decline in macro F1 performance of all models over time when evaluated on articles published in 2023, 2024, and 2025, despite being trained on data from 2019 to 2022. This trend reflects the presence of temporal distribution shift, which negatively affects the models' ability to generalize to newer data.

Specifically, traditional models such as Naive Bayes, Logistic Regression, LinearSVC, and SGDClassifier, as well as deep learning models like TextCNN, BiGRU, and TextRCNN, all exhibit performance degradation to varying extents. Among them, Transformer-based language models including mBERT, XLM-R, and PhoBERT demonstrate greater robustness against temporal changes, showing smaller drops in F1 scores compared to other approaches. PhoBERT, in particular, stands out by maintaining the highest and most stable performance across the

evaluation years, highlighting its strong capacity to capture and adapt to the evolving linguistic characteristics in Vietnamese over time.

Regarding category-level impact, although the figure does not provide explicit category-wise breakdowns, it can be inferred that categories with rapidly changing content or those influenced by emerging events are likely to suffer the most significant performance decline. This suggests the necessity of periodic model updates or retraining to preserve classification accuracy in the face of dynamic data distributions.

In summary, these findings underscore the importance of considering temporal factors when designing and deploying text classification systems, and emphasize the advantage of using powerful, deep learning-based architectures to ensure sustained and stable performance over time.

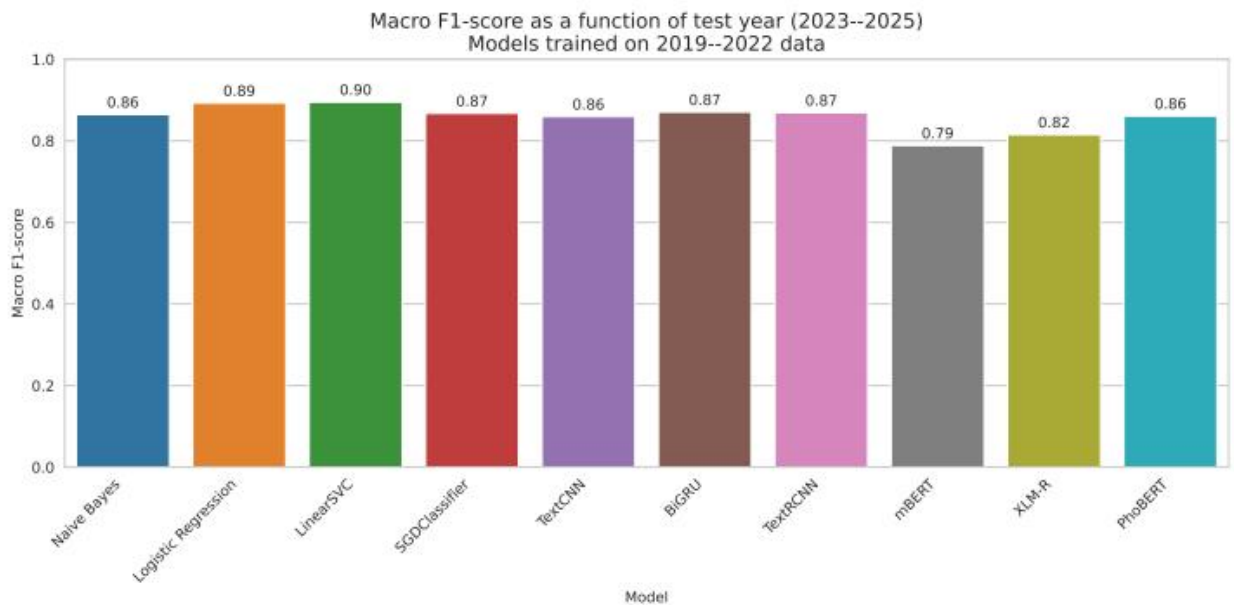


Image 13: Macro F1-score as a function of test year (2023–2025). Models trained on 2019–2022 data

RQ3: Impact of News Source on Classification Performance

To examine source-level bias, we conducted leave-one-source-out experiments in which each model was trained on articles from two outlets and evaluated on the held-out third outlet. Table 9 reports the macro F1 scores for all three leave-one-out configurations.

The results indicate that the largest performance drop occurs when the model is tested on VnExpress, with several models experiencing a significant decrease in macro F1-score. Among the traditional machine learning models, LinearSVC shows the greatest sensitivity to source bias, with drops as high as 8.07 points on VnExpress and 7.48 points on VietNamNet. In contrast, Naïve Bayes is comparatively less affected by source bias, exhibiting the smallest declines across all three sources.

For the deep learning models, BiGRU suffers the largest performance drop, especially on VnExpress (9.71 points), indicating strong source dependency. TextCNN and TextRCNN also show considerable declines but are slightly less affected than BiGRU. Among the transformer-based models, mBERT demonstrates the least sensitivity to source bias, with minimal drops across all sources, whereas PhoBERT experiences more substantial decreases, particularly on VietnamNet. Overall, these findings reveal that source bias has a notable impact on classification performance, with certain models, especially deep learning ones like BiGRU, being more vulnerable, while others, such as mBERT and Naive Bayes, show more robustness to variations in news source.

Family	Model	VnExpress	VietNam Net	Dân Trí	All sources
Traditional ML	Naive Bayes	83.83	81.35	86.31	86.36
Traditional ML	Logistic Regression	84.92	84.56	87.84	91.36
Traditional ML	LinearSVC	84.26	84.85	86.69	92.33
Traditional ML	SGDClassifier	83.50	82.53	86.54	87.03
Deep Learning	TextCNN	82.22	83.32	85.61	91.27
Deep Learning	BiGRU	82.71	83.61	86.59	92.42
Deep Learning	TextRCNN	83.12	83.32	87.13	92.16
Transformer	mBERT	77.51	74.36	78.67	78.73
Transformer	XLM-R	79.44	77.99	81.27	81.60
Transformer	PhoBERT	83.81	82.57	85.56	87.98

Table 2: Leave-one-source-out classification performance (macro F1-score)

RQ4: Impact of Training Data Scale on Classification Performance

To assess sample efficiency across model families, we trained all nine models on six subsets of the training data: 1%, 5%, 10%, 25%, 50%, and 100% of

the full training set, with category proportions preserved at each scale. Image below will presents the resulting learning curves. At small data scales (1–5% of the full training set), traditional machine learning models, particularly LinearSVC and Logistic Regression, outperform transformer-based models, achieving higher macro F1-scores. Among the transformers, PhoBERT and XLM-R show moderate performance, while mBERT lags considerably at these minimal data levels. As the training set size increases, all transformer models exhibit rapid gains in performance, with learning curves steepening between 5–25% of the data. Notably, transformer models reach a performance plateau later than classical machine learning models, indicating that they benefit more from larger datasets. The crossover point, where transformers surpass the best-performing classical models, occurs roughly around 10–25% of the training data, after which transformers consistently achieve superior macro F1-scores, demonstrating their higher capacity and scalability with increasing data size. Overall, these results highlight the differential sample efficiency: classical models excel in extremely low-data regimes, whereas transformers dominate as more training data becomes available.

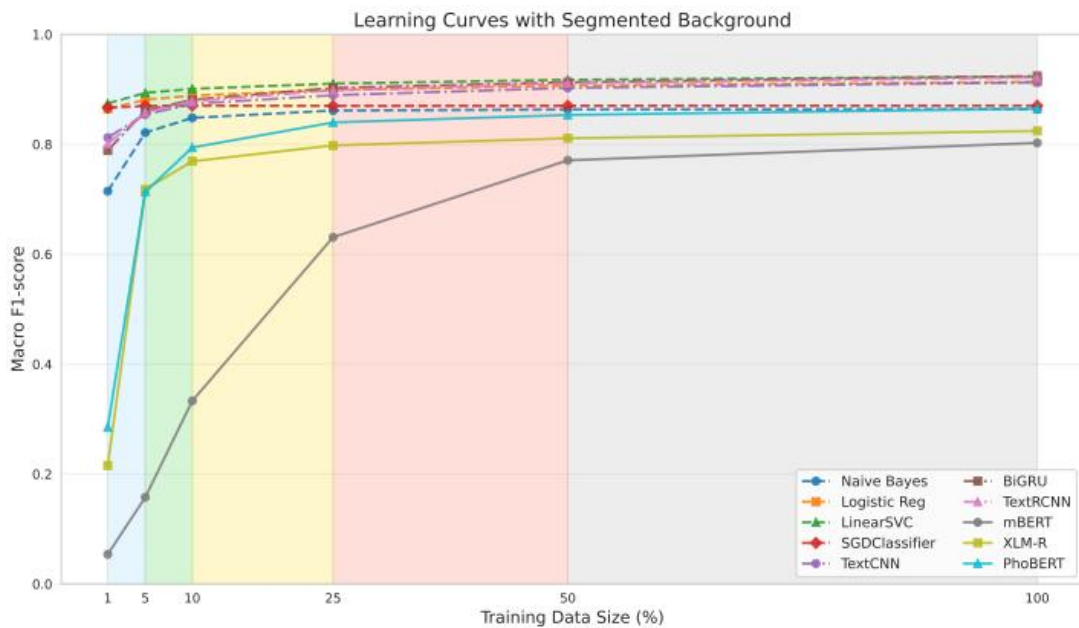


Image 14: Learning curves: macro F1-score as a function of training data size (%) for all model families.

5.5 Error Analysis

Text	Meaning	True label	Predicted label	confidence
siêu xe lamborghini invencible triệu đô đầu lăn bánh_kiến bình_luận chia_sẻ viết ban ô_tô xe_máy...	A luxury Lamborghini supercar worth millions just rolled off the assem- bly line	Automative	Automative	0.99
phát mê dàn siêu xe triệu usd đẹp tì vết đại_gia_hoàng kim_khánh_kiến bình_luận chia_sẻ viết ban ô_tô xe_máy email otoxemay vietnamnet vn nội_dung phù_hợp đăng_tải cảm_on...	Admiring a flawless collection of milliondollar supercars owned by a wealthy individual	Automative	Automative	0.99
xe thay_đổi lịch_sử giới xe xanh p rolls royce phantom hãng xerolls roycebị bmw mua lo_sợ rolls royce đi chất quý_phái quốc_hãng xe rover_nhiên thực_tế hoàn_toàn biệt_rolls royce bmw_chế xe sản_xuất phiên rolls royce phantom_định_nghĩa chuẩn_mức dòng xe siêu_danh_tiếng rolls royce audi r_giới chờ_đợi xe thể_thao_thể thay	History and evolution of luxury green cars including Rolls Royce Phan- tom and concerns about BMW	Automative	Automative	0.99

Text	Meaning	True label	Predicted label	confidence
hợp_tác nghệ_sĩ thương_hiệu quốc_tế t production_tâm huyết_đạo_diễn trần_thành trung_mắt đơn_vị đứng_dự_án quy_tụ đội_ngũ sản_xuất chuyên_nghiep liên_tục cập_nhật...	International artist collaboration and dedicated production team led by director Tran Thanh Trun	Business	Entertainment	0.99
doanh_nhan ngoai tạm_giam nguyên_huy khang_tra truy_tố xét_xử tội_lừa_đảo chiếm_đoạt tài_sản vụ_án_nhượng_vốn góp_công_ty TNHH công_nghiep trường_sinh vụ_án liên_quan	Businessman Nguyen Huy Khang detained and prosecuted for fraud and asset appropriation related to capital transfer case	Politics	Law	0.99

Table 3: Examples of Correct and Incorrect Classifications by PhoBERT Model

To better understand model performance, an error analysis was conducted based on predictions from the PhoBERT model. Table X presents examples of both correct and incorrect classifications with corresponding confidence scores.

The model performs well on categories with clear semantic boundaries, such as automotive-related articles, where domain-specific keywords help achieve highly accurate predictions with high confidence. This shows that the model effectively captures strong lexical and contextual signals.

However, misclassifications are observed in categories with overlapping content, such as business and entertainment or politics and law. These errors occur due to shared vocabulary and similar contexts, making it difficult for the model to distinguish between closely related classes.

Additionally, some incorrect predictions are associated with high confidence scores, indicating that the model can be overconfident in ambiguous cases. Data-related factors, such as class imbalance and noisy or overlapping labels, may also contribute to these errors.

Overall, while PhoBERT demonstrates strong performance, it still faces challenges in fine-grained classification, highlighting areas for future improvement.

CHƯƠNG 6. CONCLUSION

6.1 Conclusion

In this study, we developed a comprehensive framework for Vietnamese news classification by combining traditional machine learning, deep learning, and Transformer-based approaches.

The experimental results show that model architecture has a significant impact on classification performance. Among all models, BiGRU achieved the best overall performance, while LinearSVC also demonstrated competitive results when combined with TF-IDF features. In contrast, Transformer-based models such as mBERT and XLM-RoBERTa did not outperform other approaches under the current experimental conditions.

Additionally, the study highlights the influence of dataset characteristics, including data scale, news sources, and temporal variations, on model performance. The results indicate that increasing the amount of training data improves classification effectiveness, while variations in news sources and publication time have a relatively limited impact.

Overall, this work demonstrates that both traditional and deep learning models remain strong baselines for text classification tasks, and that careful design of data processing and feature representation is crucial for achieving high performance.

6.2 Limitations of the Study

Despite achieving promising results, this study has several limitations.

First, the dataset was collected from a limited number of Vietnamese news sources, which may not fully represent the diversity of writing styles and topics across all online media. This could affect the generalizability of the models to unseen data from other sources.

Second, although multiple model architectures were evaluated, the training configurations were relatively constrained. In particular, Transformer-based models such as mBERT and XLM-RoBERTa were trained with a limited number of epochs and computational resources, which may have prevented them from reaching optimal performance.

Third, the study primarily focused on textual content and did not incorporate additional features such as metadata (e.g., author, publication time) or external knowledge sources, which could potentially improve classification accuracy.

Finally, although the dataset is relatively large, further scaling the data and exploring more advanced preprocessing or representation techniques may lead to better performance, especially for deep learning and Transformer-based models.

6.3 Future Work

To further improve the effectiveness and applicability of the proposed system, several directions for future research can be considered.

First, more advanced language models such as transformer-based architectures (PhoBERT or multilingual BERT) could be explored to better capture semantic information from Vietnamese symptom descriptions. These models are capable of understanding contextual relationships beyond word frequency, potentially leading to improved diagnostic accuracy.

Second, incorporating additional clinical features—such as gender, duration of symptoms, medical history, or laboratory test results—would allow the model to make more informed predictions and better reflect real-world diagnostic processes.

Third, future work could extend the current single-label classification framework to a multi-label setting, enabling the system to predict multiple possible diseases simultaneously. This would better align the model with real clinical scenarios, where comorbidities are common.

Finally, deploying the model as a web-based or mobile application could facilitate real-world testing and user feedback. Such deployment would also enable continuous data collection, allowing the model to be retrained and improved over time.

6.4 Final Remarks

Future work can focus on several directions to further improve the performance of Vietnamese news classification models.

First, expanding the dataset by collecting data from more diverse news sources and increasing the data scale may help improve model generalization and robustness.

Second, more advanced fine-tuning strategies can be explored for Transformer-based models such as mBERT and XLM-RoBERTa, including longer training, better hyperparameter tuning, and improved optimization techniques.

Third, incorporating additional features such as metadata (e.g., publication time, author information) or external knowledge sources may enhance the model's ability to capture richer contextual information.

Finally, future research can investigate more advanced architectures or hybrid approaches that combine the strengths of traditional machine learning and deep learning methods to achieve better classification performance.

REFERENCES

- Vu, T., Nguyen, D. Q., Nguyen, D. Q., Dras, M., & Johnson, M. (2018). VnCoreNLP: A Vietnamese natural language processing toolkit. *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Demonstrations*, 56–60.
- Zhang, X., Zhao, J., & LeCun, Y. (2015). Character-level convolutional networks for text classification. *Advances in Neural Information Processing Systems*, 28, 649–657.
- Lewis, D. D., Yang, Y., Rose, T. G., & Li, F. (2004). RCV1: A new benchmark collection for text categorization research. *Journal of Machine Learning Research*, 5, 361–397.
- Wu, F., Qiao, Y., Chen, J.-H., Wu, C., Qi, T., Lian, J., Liu, D., Xie, X., Gao, J., Wu, W., & Zhou, M. (2020). MIND: A large-scale dataset for news recommendation. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 3597–3606.
- McCallum, A., & Nigam, K. (1998). A comparison of event models for naive Bayes text classification. *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Genkin, A., Lewis, D. D., & Madigan, D. (2007). Large-scale Bayesian logistic regression for text categorization. *Technometrics*, 49(3), 291–304.
- Joachims, T. (1998). Text categorization with support vector machines: Learning with many relevant features. *Machine Learning: ECML-98*, 137–142.
- Kim, Y. (2014). Convolutional neural networks for sentence classification. *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, 1746–1751.

Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder–decoder for statistical machine translation. *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, 1724–1734.

Schuster, M., & Paliwal, K. (1997). Bidirectional recurrent neural networks. *IEEE Transactions on Signal Processing*, 45(11), 2673–2681.

Lai, S., Xu, L., Liu, K., & Zhao, J. (2015). Recurrent convolutional neural networks for text classification. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2267–2273.

Nguyen, D. Q., & Nguyen, A. T. (2020). PhoBERT: Pre-trained language models for Vietnamese. *Findings of the Association for Computational Linguistics*, 1037–1042.

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *Proceedings of NAACL-HLT*, 4171–4186.

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., & Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. *Proceedings of ACL*, 8440–8451.

Hoang, V. C. D., Dinh, D., Nguyen, N. L., & Ngo, H. Q. (2007). A comparative study on Vietnamese text classification methods. *Proceedings of the IEEE International Conference on Research, Innovation and Vision for the Future*, 267–273.

Nguyen, K. V., Nguyen, V. D., Nguyen, P. X. V., Truong, T. T. H., & Nguyen, N. L.-T. (2018). UIT-VSFC: Vietnamese students’ feedback corpus for sentiment

analysis. *Proceedings of the International Conference on Knowledge and Systems Engineering*, 19–24.

Vinh, T., & Ha, K. (2021). Vietnamese news articles classification using neural networks. *Journal of Advances in Information Technology*, 12.

Cui, M., Bai, R., Lu, Z., Li, X., Aickelin, U., & Ge, P. (2019). Regular expression based medical text classification using constructive heuristic approach. *IEEE Access*, 7, 147892–147904.

Wang, J., Li, Y., Shan, J., Bao, J., Zong, C., & Zhao, L. (2019). Large-scale text classification using scope-based convolutional neural network. *IEEE Access*, 7, 171548–171558.

Gong, J., Teng, Z., Teng, Q., Zhang, H., Du, L., Chen, S., Bhuiyan, M. Z. A., Li, J., Liu, M., & Ma, H. (2020). Hierarchical graph transformer-based deep learning model for large-scale multi-label text classification. *IEEE Access*, 8, 30885–30896.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.

Lazaridou, A., Kuncoro, A., Gribovskaya, E., Agrawal, D., Liška, A., Terzi, T., Gimenez, M., d’Autume, C. M., Kocisky, T., Ruder, S., Yogatama, D., Cao, K., Young, S., & Blunsom, P. (2021). Mind the gap: Assessing temporal generalization in neural language models. *Advances in Neural Information Processing Systems*.

Bi, K., Jha, R., Croft, B., & Celikyilmaz, A. (2021). AREDSUM: Adaptive redundancy-aware iterative sentence ranking for extractive document summarization. *Proceedings of EACL*, 281–291.

Duong, H.-T., & Hoang, V. T. (2019). A survey on multiple classifiers for Vietnamese news classification. *Proceedings of the International Conference on Knowledge and Smart Technology*, 23–28.

Tran, K. Q., Trinh, P. N., Tran, K. N.-A., Le, A. T.-H., Ha, L. V., & Nguyen, K. V. (2021). An empirical investigation of online news classification on a large-scale Vietnamese dataset. *Frontiers in Artificial Intelligence and Applications*, 337, 367–379.

Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., Cistac, P., Rault, T., Louf, R., Funtowicz, M., Davison, J., Shleifer, S., von Platen, P., Ma, C., Jernite, Y., Plu, J., Xu, C., Le Scao, T., Gugger, S., ... Rush, A. M. (2020). Transformers: State-of-the-art natural language processing. *Proceedings of EMNLP: System Demonstrations*, 38–45.